

Robust State Space Filtering Under Incremental Model Perturbations Subject to a Relative Entropy Tolerance

Bernard C. Levy, *Fellow, IEEE*, and Ramine Nikoukhah

Abstract—This paper considers robust filtering for a nominal Gaussian state-space model, when a relative entropy tolerance is applied to each time increment of a dynamical model. The problem is formulated as a dynamic minimax game where the maximizer adopts a myopic strategy. This game is shown to admit a saddle point whose structure is characterized by applying and extending results presented earlier in “Robust least-squares estimation with a relative entropy constraint” (B. C. Levy and R. Nikoukhah, *IEEE Trans. Inf. Theory*, vol. 50, no. 1, 89–104, Jan. 2004) for static least-squares estimation. The resulting minimax filter takes the form of a risk-sensitive filter with a time varying risk sensitivity parameter, which depends on the tolerance bound applied to the model dynamics and observations at the corresponding time index. The least-favorable model is constructed and used to evaluate the performance of alternative filters. Simulations comparing the proposed risk-sensitive filter to a standard Kalman filter show a significant performance advantage when applied to the least-favorable model, and only a small performance loss for the nominal model.

Index Terms—Commitment, dynamic minimax game, least-favorable model, myopic strategy, relative entropy, risk-sensitive filtering, robust filtering.

I. INTRODUCTION

SOON after the introduction of Wiener and Kalman filters, it was recognized that these filters were vulnerable to modeling errors, in the form of either parasitic signals or perturbations of the system dynamics. Various approaches were proposed over the last 35 years to construct filters with a guaranteed level of immunity to modelling uncertainties. Drawing from the framework developed by Huber for robust statistics [2], Kassam, Poor and their collaborators proposed an approach [3]–[5] where the optimum filter is selected by solving a minimax problem. In this approach, the set of possible system models is described by a neighborhood centered about the nominal model, and two players affront each other. One player (say, nature) selects the least-favorable model in the allowable neighborhood and the

other player designs the optimum filter for the least-favorable model. While minimax filtering is conceptually simple, its implementation can be very difficult, since it depends on the specification of the allowable neighborhood and of the loss function to be minimized. After some early success in the design of Wiener filters for neighborhoods specified by ϵ -contamination models or power spectral bands, progress stalled gradually and researchers began considering alternative robust filtering approaches. The 1980s saw the development of a new class of robust filters based on the minimization of risk-sensitive and H^∞ performance criteria [6]–[10]. This approach seeks to avoid large errors, even if these errors are unlikely based on the nominal model. For example, risk-sensitive filters replace the standard quadratic loss function of least-squares filtering by an exponential quadratic function, which of course penalizes severely large errors. However, errors in the model dynamics are not introduced explicitly in H^∞ and risk-sensitive filtering, and the growing awareness of the importance of such errors prompted a number of researchers in the early 2000s [11]–[13] to revive the minimax filtering viewpoint, but in a context where modeling errors are described in terms of norms for state-space dynamics perturbations. The present paper, which is a continuation of [1], can be viewed as part of a larger effort initiated by Hansen and Sargent [14]–[16] and other researchers [17], [18] which is aimed at reinterpreting risk-sensitive filtering from a minimax viewpoint. In this context, modelling uncertainties are described by specifying a tolerance for the relative entropy between the actual system and the nominal model. To a fixed tolerance level describing the modeler’s confidence in the nominal model corresponds a ball of possible models for which it is then possible to apply the minimax filtering approach proposed by Kassam and Poor.

The minimax formulation of robust filtering based on a relative entropy constraint has several attractive features. First, relative entropy is a natural measure of model mismatch which is commonly used by statisticians for fitting statistical models by using techniques such as the expectation maximization iteration [19]. In addition, it turns out that minimax Wiener and Kalman filtering problems with a relative entropy constraint admit solutions [1], [14], [15], [18] in the form of risk-sensitive filters, thus providing a new interpretation for such filters. The main difference between earlier works and the present paper is that, instead of placing a single relative entropy constraint on the entire system model, we apply a separate constraint to each time increment of the model. This approach, which is closer to the one advocated in [12], [13] is based on the following consideration.

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B. Levy is with the Department of Electrical and Computer Engineering, University of California, Davis, CA 95616 USA (e-mail: bclevy@ucdavis.edu).

R. Nikoukhah is with the Institut de Recherche en Informatique et Automatique (INRIA), Domaine de Voluceau, Rocquencourt, 78153 Le Chesnay Cedex France (e-mail: ramine.nikoukhah@inria.fr).

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When a single relative entropy constraint is placed on the complete system model, the maximizing player has the opportunity to allocate almost all of its mismatch budget to a single element of the model most susceptible to uncertainties. But this strategy may lead to overly pessimistic conclusions, since in practice modelers allocate the same level of effort to modeling each time component of the system. Thus, it probably makes better sense to specify a fixed uncertainty tolerance for each model increment, instead of a single bound for the overall model.

The analysis presented relies in part on applying and extending static least-squares estimation results derived in [1] for nominal Gaussian models. These results are reviewed in Section II. In Section III, the robust state-space filtering problem with an incremental relative entropy constraint is formulated as a dynamic game where the maximizer adopts a myopic strategy whose goal is to maximize the mean-square estimation error at the current time only. The existence of a saddle point is established in Section IV, where the least-favorable model specifying the saddle point is characterized by extending a Lemma of [1] to the dynamic case. A careful examination of the least-favorable model structure allows the transformation of the dynamic estimation game into an equivalent static one, to which the result of [1] become applicable. The robust filter that we obtain is a risk-sensitive filter, but with a time-varying risk-sensitivity parameter, representing the inverse of the Lagrange multiplier associated to the model component constraint for the corresponding time period. The least-favorable state-space model for which the robust filter is optimal is constructed in Section V. This model allows evaluation studies comparing the performance of the minimax filter with that of other filters, such as the standard Kalman filter. Simulations are presented in Section VI which illustrate the dependence of the filter performance on the relative entropy tolerance applied to each model time component. The robust filter is compared to the ordinary Kalman filter by examining their respective performances for both the nominal and least-favorable systems. Finally, some conclusions are presented in Section VII.

II. ROBUST STATIC ESTIMATION

We start by reviewing a robust static estimation result derived in [1], since its extension to the dynamic case is the basis for the robust filtering scheme we propose. Let

$$z = \begin{bmatrix} x \\ y \end{bmatrix} \quad (1)$$

be a random vector of $\mathbb{R}^{n+p}$, where $x \in \mathbb{R}^n$ is a vector to be estimated, and $y \in \mathbb{R}^p$ is an observed vector. The nominal and actual probability densities of z are denoted, respectively, as $f(z)$ and $\tilde{f}(z)$. The deviation of $\tilde{f}$ from f is measured by the relative entropy (Kullback–Leibler divergence)

$$D(\tilde{f}, f) = \int_{\mathbb{R}^{n+p}} \tilde{f}(z) \ln \left(\frac{\tilde{f}(z)}{f(z)} \right) dz. \quad (2)$$

The relative entropy is not a distance, since it is not symmetric and does not satisfy the triangle inequality. However it has the property that $D(\tilde{f}, f) \geq 0$ with equality if and only if $\tilde{f} = f$.

Furthermore, since the function $\theta(\ell) = \ell \ln(\ell)$ is convex for $0 \leq \ell < \infty$, $D(\tilde{f}, f)$ is a convex function of $\tilde{f}$. For a fixed tolerance $c > 0$, if $\mathcal{F}$ denotes the class of probability densities over $\mathbb{R}^{n+p}$, this ensures that the “ball”

$$\mathcal{B} = \{ \tilde{f} \in \mathcal{F} : D(\tilde{f}, f) \leq c \} \quad (3)$$

of densities $\tilde{f}$ located within a divergence tolerance c of the nominal density f is a closed convex set. $\mathcal{B}$ represents the set of all possible true densities of random vector z consistent with the allowed mismodeling tolerance.

Throughout this paper we shall adopt a minimax viewpoint of robustness similar to [2], [14], where whenever we seek to design an estimator minimizing an appropriately selected loss function, a hostile player, say “nature,” conspires to select the worst possible model in the allowed set $\mathcal{B}$. This approach is rather conservative, but the degree of conservativeness can be controlled by appropriate selection of the tolerance parameter c , which ensures that an adequate balance between performance and robustness is reached.

In this paper, we shall use the mean-square error (scaled by 1/2)

$$\begin{aligned} J(\tilde{f}, g) &= \frac{1}{2} \tilde{E} \left[\|x - g(y)\|^2 \right] \\ &= \frac{1}{2} \int_{\mathbb{R}^{n+p}} \|x - g(y)\|^2 \tilde{f}(z) dz \end{aligned} \quad (4)$$

to evaluate the performance of an estimator $\hat{x} = g(y)$ of x based on observation y . In (4), if v denotes a vector of $\mathbb{R}^n$ with entries v_i

$$\|v\| = (v^T v)^{\frac{1}{2}} = \left(\sum_i v_i^2 \right)^{\frac{1}{2}}$$

denotes the usual Euclidean vector norm. Let $\mathcal{G}$ denote the class of estimators such that $\tilde{E}[\hat{x}^2]$ is finite for all $\tilde{f} \in \mathcal{B}$. Then the optimal robust estimator solves the minimax problem

$$\min_{g \in \mathcal{G}} \max_{\tilde{f} \in \mathcal{B}} J(\tilde{f}, g). \quad (5)$$

Since the functional $J(\tilde{f}, g)$ is quadratic in g , and thus convex, and linear in $\tilde{f}$, and thus concave, a saddle-point $(\tilde{f}_0, g_0)$ of minimax problem (5) exists, so that

$$J(\tilde{f}, g_0) \leq J(\tilde{f}_0, g_0) \leq J(\tilde{f}_0, g). \quad (6)$$

However, characterizing precisely this saddle-point is difficult, except when the nominal density is Gaussian, i.e.,

$$f(z) \sim N(m_z, K_z) \quad (7)$$

where in conformity with partition (1) of z , the mean vector m_z and covariance matrix K_z admit the partitions

$$m_z = \begin{bmatrix} m_x \\ m_y \end{bmatrix}, \quad K_z = \begin{bmatrix} K_x & K_{xy} \\ K_{yx} & K_y \end{bmatrix}.$$

Then it was shown in Theorem 1 of [1] (see also [14, Sec. 7.3] for an equivalent result derived from a stochastic game theory

perspective) that the saddle point of minimax problem (5) admits the following structure.

Theorem 1: If f admits the Gaussian distribution (7), the least-favorable density $\tilde{f}_0$ is also Gaussian with distribution

$$\tilde{f}_0 \sim N(m_z, \tilde{K}_z) \quad (8)$$

where the covariance matrix

$$\tilde{K}_z = \begin{bmatrix} \tilde{K}_x & K_{xy} \\ K_{yx} & K_y \end{bmatrix} \quad (9)$$

is obtained by perturbing only the covariance matrix of x , leaving the cross- and co-variance matrices K_{xy} and K_y unchanged. Accordingly, the robust estimator

$$\hat{x} = g_0(y) = m_x + K_{xy}K_y^{-1}(y - m_y) \quad (10)$$

coincides with the usual least-squares estimator for nominal density f . The perturbed covariance matrix $\tilde{K}_x$ can be evaluated as follows. Let

$$\begin{aligned} P &= K_x - K_{xy}K_y^{-1}K_{yx} \\ \tilde{P} &= \tilde{K}_x - K_{xy}K_y^{-1}K_{yx} \end{aligned} \quad (11)$$

denote the nominal and least-favorable error covariance matrices of x given y . Then

$$\tilde{P}^{-1} = P^{-1} - \lambda^{-1}I_n \quad (12)$$

where λ denotes the Lagrange multiplier corresponding to constraint $D(\tilde{f}, f) \leq c$. Note that to ensure that $\tilde{P}$ is a positive definite matrix, we must have $\lambda > r(P)$, where $r(P)$ denotes the spectral radius (the largest eigenvalue) of P .

To explain precisely how λ is selected to ensure that the Karush–Kuhn–Tucker (KKT) condition

$$\lambda (c - D(\tilde{f}_0, f)) = 0 \quad (13)$$

holds, observe first that for two Gaussian densities $f \sim N(m_z, K_z)$ and $\tilde{f} \sim N(\tilde{m}_z, \tilde{K}_z)$, the relative entropy can be expressed as [20]

$$D(\tilde{f}, f) = \frac{1}{2} \left[\|\Delta m_z\|_{K_z^{-1}}^2 + \text{tr} \left(K_z^{-1} \tilde{K}_z - I_{n+p} \right) - \ln \det \left(K_z^{-1} \tilde{K}_z \right) \right] \quad (14)$$

where $\Delta m_z = \tilde{m}_z - m_z$ and $\|v\|_{K^{-1}} \triangleq (v^T K^{-1} v)^{1/2}$. Then for the nominal and least-favorable densities specified by (7) and (8), (9), we have $\Delta m_z = 0$ and

$$\begin{aligned} K_z &= \begin{bmatrix} I_n & G_0 \\ 0 & I_p \end{bmatrix} \begin{bmatrix} P & 0 \\ 0 & K_y \end{bmatrix} \begin{bmatrix} I_n & 0 \\ G_0^T & I_p \end{bmatrix} \\ \tilde{K}_z &= \begin{bmatrix} I_n & G_0 \\ 0 & I_p \end{bmatrix} \begin{bmatrix} \tilde{P} & 0 \\ 0 & K_y \end{bmatrix} \begin{bmatrix} I_n & 0 \\ G_0^T & I_p \end{bmatrix} \end{aligned}$$

where $G_0 \triangleq K_{xy}K_y^{-1}$ denotes the gain matrix of the optimal estimator (10). Then after simple algebraic manipulations, we find

$$D(\tilde{f}_0, f) = \frac{1}{2} \left[\text{tr}(\tilde{P}P^{-1} - I_n) - \ln \det(\tilde{P}P^{-1}) \right]. \quad (15)$$

Substituting (12) gives

$$\begin{aligned} \gamma(\lambda) &\triangleq D(\tilde{f}_0, f) \\ &= \frac{1}{2} \left[\text{tr} \left((I_n - \lambda^{-1}P)^{-1} - I_n \right) + \ln \det(I_n - \lambda^{-1}P) \right] \end{aligned} \quad (16)$$

where $\gamma(\lambda)$ is differentiable over $(r(P), \infty)$. By using the matrix differentiation identities [21, Chap. 8]

$$\begin{aligned} \frac{d}{d\lambda} \ln \det M(\lambda) &= \text{tr} \left[M^{-1} \frac{dM}{d\lambda} \right] \\ \frac{d}{d\lambda} M^{-1}(\lambda) &= -M^{-1} \frac{dM}{d\lambda} M^{-1} \end{aligned}$$

for a square invertible matrix function $M(\lambda)$, we find

$$\frac{d\gamma}{d\lambda} = -\text{tr} \left[(I - \lambda^{-1}P)^{-1} \lambda^{-3} P^2 (I - \lambda^{-1}P)^{-1} \right] < 0 \quad (17)$$

so that $\gamma(\lambda)$ is monotone decreasing over $(r(P), \infty)$. Since

$$\lim_{\lambda \rightarrow r(P)} \gamma(\lambda) = +\infty, \quad \lim_{\lambda \rightarrow \infty} \gamma(\lambda) = 0$$

this ensures that for an arbitrary tolerance $c > 0$, there exists a unique $\lambda > r(P)$ such that $\gamma(\lambda) = c$.

For the case where the nominal density f is non-Gaussian, some results characterizing the solution of the minimax problem (5) were described recently in [22].

III. ROBUST FILTERING VIEWED AS A DYNAMIC GAME

We consider a robust state-space filtering problem for processes described by a nominal Gauss–Markov state-space model of the form

$$x_{t+1} = A_t x_t + B_t v_t \quad (18)$$

$$y_t = C_t x_t + D_t v_t \quad (19)$$

where $v_t \in \mathbb{R}^m$ is a WGN with unit variance, i.e.,

$$E[v_t v_s^T] = I_m \delta(t - s) \quad (20)$$

where

$$\delta(r) = \begin{cases} 1 & r = 0 \\ 0 & r \neq 0 \end{cases}$$

denotes the Kronecker delta function. The noise v_t is assumed to be independent of the initial state, whose nominal distribution is given by

$$f_0(x_0) \sim N(m_0, P_0). \quad (21)$$

Let

$$z_t \triangleq \begin{bmatrix} x_{t+1} \\ y_t \end{bmatrix}$$

The model (18)–(20) can be viewed as specifying the nominal conditional density

$$\phi_t(z_t|x_t) \sim N \left(\begin{bmatrix} A_t \\ C_t \end{bmatrix} x_t, \begin{bmatrix} B_t \\ D_t \end{bmatrix} \begin{bmatrix} B_t^T & D_t^T \end{bmatrix} \right). \quad (22)$$

of z_t given x_t . We assume that the noise v_t affects all components of the dynamics (18) and observations (19), so that the covariance matrix

$$K_{z_t|x_t} = \begin{bmatrix} B_t \\ D_t \end{bmatrix} [B_t^T \quad D_t^T] \quad (23)$$

is positive definite. To interpret this assumption, observe that in general, state-space models of the form (18), (19) are formed by a mixture of noisy and deterministic linear relations (see for example the decomposition employed in [23]). This means that the resulting conditional densities are concentrated on lower dimensional subspaces of $\mathbb{R}^{n+p}$. As soon as these subspaces are slightly perturbed, it is possible to discriminate perfectly between the nominal and perturbed models, i.e., the relative entropy of the two models is infinite (the corresponding probability measures are not absolutely continuous with respect to each other). Accordingly, when the relative entropy is used to measure the proximity of statistical models, all deterministic relations between dynamic variables or observations are interpreted as immune from uncertainty, and only relations where noise is already present can be perturbed. Since this limitation is rather unsatisfactory, it is convenient to assume, as in earlier robust filtering studies [12], [17], [18], that the noise v_t affects all components of the dynamics and observations, possibly with a very small variance for relations which are viewed as almost certain.

In this case, since the matrix

$$\Gamma_t \triangleq \begin{bmatrix} B_t \\ D_t \end{bmatrix}$$

has full row rank, we can assume without loss of generality that Γ_t is square and invertible, so that $m = n + p$. Otherwise, we could compress the column space of Γ_t and remove noises which do not affect the model (18), (19).

Over a finite interval $0 \leq t \leq T$, the joint nominal probability density of

$$X_{T+1} = \begin{bmatrix} x_0 \\ \vdots \\ x_t \\ \vdots \\ x_{T+1} \end{bmatrix} \quad \text{and} \quad Y_T = \begin{bmatrix} y_0 \\ \vdots \\ y_t \\ \vdots \\ y_T \end{bmatrix}$$

can be expressed as

$$f(X_{T+1}, Y_T) = f_0(x_0) \prod_{t=0}^T \phi_t(z_t|x_t) \quad (24)$$

where the initial and the combined state transition and observation densities are given by (21) and (22). Assume that the true probability density of X_{T+1} and Y_T admits a similar Markov structure of the form

$$\tilde{f}(X_{T+1}, Y_T) = \tilde{f}_0(x_0) \prod_{t=0}^T \tilde{\phi}_t(z_t|x_t). \quad (25)$$

By taking the expectation of

$$\ln \left(\frac{\tilde{f}(X_{T+1}, Y_T)}{f(X_{T+1}, Y_T)} \right) = \ln \left(\frac{\tilde{f}_0(x_0)}{f_0(x_0)} \right) + \sum_{t=0}^T \ln \left(\frac{\tilde{\phi}_t(z_t|x_t)}{\phi_t(z_t|x_t)} \right)$$

with respect to $\tilde{f}(X_{T+1}, Y_T)$, we find that the relative entropy between $\tilde{f}(X_{T+1}, Y_T)$ and $f(X_{T+1}, Y_T)$ satisfies the chain rule

$$D(\tilde{f}, f) = D(\tilde{f}_0, f_0) + \sum_{t=0}^T D(\tilde{\phi}_t, \phi_t) \quad (26)$$

with

$$\begin{aligned} D(\tilde{\phi}_t, \phi_t) &= \tilde{E} \left[\ln \left(\frac{\tilde{\phi}_t(z_t|x_t)}{\phi_t(z_t|x_t)} \right) \right] \\ &= \iint \tilde{\phi}_t(z_t|x_t) \tilde{f}_t(x_t) \ln \left(\frac{\tilde{\phi}_t(z_t|x_t)}{\phi_t(z_t|x_t)} \right) dz_t dx_t \end{aligned} \quad (27)$$

where $\tilde{f}_t(x_t)$ denotes the true marginal density of x_t . Up to this point, most results on robust Kalman filtering with a relative entropy constraint have been obtained by considering a fixed interval and applying a single constraint to the relative entropy $D(\tilde{f}, f)$ of the true and nominal probability densities of the state and observation sequences over the whole interval. To treat the robust filtering problem over an infinite horizon, one approach consists of dividing the divergence over a finite interval by the length T of the interval, and letting T tend to infinity, assuming this sequence has a limit. This is the case for stationary Gaussian processes, since in this case the limit is the Itakura–Saito spectral distortion measure [1, Sec. 3]. Alternatively, it is also possible to apply a discount factor [15] to future additive terms appearing in the chain rule decomposition (26). However, one potential weakness of applying a single divergence constraint to the filtering problem over a finite or infinite interval is that it allows the maximizer (nature) to identify the moment where the dynamic model (18)–(20) is most susceptible to distortions and to allocate most of the distortion budget specified by the tolerance parameter c to this single element of the model. If the modeler exercises the same level of effort to characterize each time component of the model (18)–(20), it may be more appropriate to specify separate modelling tolerances for each time step of the transition density (22). Such a viewpoint has in fact been adopted widely [11]–[13] in the robust state-space filtering literature, except that in these earlier studies the tolerance is usually expressed in terms of matrix bounds involving the matrices A_t , B_t , C_t , and D_t parametrizing the state dynamics and observations. In contrast with these earlier studies, we use here the relative entropy $D(\tilde{\phi}_t, \phi_t)$ between the true and nominal transition and observation densities $\tilde{\phi}_t(z_t|x_t)$ and $\phi_t(z_t|x_t)$ at time t to measure modeling errors.

The expression (27) for the relative entropy raises immediately the issue of how to choose the probability density $\tilde{f}_t(x_t)$ used to evaluate the divergence. We assume that, like the estimating player, at time t the maximizer has access to the observations $\{y(s), 0 \leq s \leq t-1\}$ collected prior to this point. In addition, it is reasonable to hold the maximizer to the same Markov structure (specified by (25)) as the estimating player. Therefore, the maximizer is required to commit to all the least-favorable

model components $\tilde{\phi}_s(z_s|x_s)$ with $0 \leq s \leq t-1$ generated at earlier stages of its minimax game with the estimating player. Using the terminology coined in [14], [15], the maximizer operates “under commitment.” Thus if Y_{t-1} denotes the vector formed by the observations $\{y_s, 0 \leq s \leq t-1\}$, we use the conditional density $\tilde{f}_t(x_t|Y_{t-1})$ based on the least favorable model and the given observations prior to time t , to evaluate the divergence (27) between the true and nominal transition and observation densities. The model mismatch tolerance can therefore be expressed as

$$\tilde{E} \left[\ln \left(\frac{\tilde{\phi}_t(z_t|x_t)}{\phi_t(z_t|x_t)} \right) \middle| Y_{t-1} \right] \leq c_t \quad (28)$$

where c_t denotes the tolerance parameter for the time t component of the model, with

$$\begin{aligned} & \tilde{E} \left[\ln \left(\frac{\tilde{\phi}_t(z_t|x_t)}{\phi_t(z_t|x_t)} \right) \middle| Y_{t-1} \right] \\ &= \iint \tilde{\phi}_t(z_t|x_t) \tilde{f}_t(x_t|Y_{t-1}) \ln \left(\frac{\tilde{\phi}_t(z_t|x_t)}{\phi_t(z_t|x_t)} \right) dz_t dx_t. \end{aligned} \quad (29)$$

Let $\mathcal{B}_t$ denote the convex ball of functions $\tilde{\phi}_t(z_t|x_t)$ satisfying inequality (28). If $\mathcal{G}_t$ denotes the class of estimators with finite second-order moments with respect to all densities $\tilde{\phi}_t(z_t|x_t)\tilde{f}_t(x_t|Y_{t-1})$ such that $\tilde{\phi}_t(z_t|x_t) \in \mathcal{B}_t$, the dynamic minimax game we consider can be expressed as

$$\min_{g_t \in \mathcal{G}_t} \max_{\tilde{\phi}_t \in \mathcal{B}_t} J_t(\tilde{\phi}_t, g_t) \quad (30)$$

where

$$\begin{aligned} J_t(\tilde{\phi}_t, g_t) &= \frac{1}{2} \tilde{E} \left[\|x_{t+1} - g_t(y_t)\|^2 | Y_{t-1} \right] \\ &= \frac{1}{2} \iint \|x_{t+1} - g_t(y_t)\|^2 \tilde{\phi}_t(z_t|x_t) \\ &\quad \times \tilde{f}_t(x_t|Y_{t-1}) dz_t dx_t \end{aligned} \quad (31)$$

denotes the mean-square error of estimate $\hat{x}_{t+1} = g_t(y_t)$ of x_{t+1} evaluated with respect to the true probability density of z_t . Note that since $\hat{x}_{t+1}$ is a function of Y_t , it depends not only on y_t , but also on earlier observations, but this dependency is suppressed to simplify notations. Note that by taking iterated expectations, we have

$$\begin{aligned} \tilde{E} \left[\|x_{t+1} - \hat{x}_{t+1}\|^2 \right] &= \tilde{E} \left[\tilde{E} \left[\|x_{t+1} - \hat{x}_{t+1}\|^2 | Y_{t-1} \right] \right] \\ &= \int J_t(\tilde{\phi}_t, g_t) \tilde{f}_{t-1}(Y_{t-1}) dY_{t-1} \end{aligned} \quad (32)$$

where $\tilde{f}_{t-1}(Y_{t-1})$ represents the least favorable density of observation vector Y_{t-1} based on the least-favorable model increments selected by the maximizer up to time $t-1$. Since this density is nonnegative and independent of both $\tilde{\phi}_t$ and g_t , the game (30) is equivalent to

$$\min_{g_t \in \mathcal{G}_t} \max_{\tilde{\phi}_t \in \mathcal{B}_t} \tilde{E} \left[\|x_{t+1} - \hat{x}_{t+1}\|^2 \right]. \quad (33)$$

This indicates that the robust least-squares filtering problem we consider is local in the sense that the estimator and the maximizer focus respectively on minimizing and maximizing the mean-square estimation error at the current time. In other words, in addition to being committed to past least-favorable model increments $\tilde{\phi}_s$ with $0 \leq s \leq t-1$ it has already selected, the maximizer is confined to a *myopic strategy*, where $\tilde{\phi}_t$ is selected exclusively to maximize the mean-square estimation error at time t . By doing so, the maximizer foregoes the possibility of trading off a lesser increase in the mean-square error at the current time against larger increases in the future.

If we compare the dynamic estimation game (30) and its static counterpart (5), we see that the two problems are similar, but the dynamic game (30) includes a conditioning operation with respect to the prior state x_t , combined with an averaging operation with respect to $\tilde{f}_t(x_t|Y_{t-1})$. Thus, Lemma 1 and Theorem 1 of [1] need to be extended slightly to accommodate these differences. Before proceeding with this task it is worth pointing out that we do not require that $\tilde{\phi}_t(z_t|x_t)$ should be a conditional probability density for each x_t . It is only required that the product $\tilde{\phi}_t(z_t|x_t)\tilde{f}_t(x_t|Y_{t-1})$ should be a probability density for

$$\begin{bmatrix} z_t \\ x_t \end{bmatrix} = \begin{bmatrix} x_{t+1} \\ y_t \\ x_t \end{bmatrix}$$

so that

$$I_t(\tilde{\phi}_t) \triangleq \iint \tilde{\phi}_t(z_t|x_t) \tilde{f}_t(x_t|Y_{t-1}) dz_t dx_t = 1. \quad (34)$$

To put it another way, the maximizer's commitment to earlier components of the least-favorable model is only of an *a priori* nature, since the *a posteriori* marginal density of x_t specified by the joint density $\tilde{\phi}_t(z_t|x_t)\tilde{f}_t(x_t|Y_{t-1})$ is not required to coincide with the *a priori* density $\tilde{f}_t(x_t|Y_{t-1})$. However, by integrating out z_t , the resulting *a posteriori* density will be of the form $\psi_t(x_t)\tilde{f}_t(x_t|Y_{t-1})$ where $\psi_t(x_t)$ does not depend on Y_{t-1} .

Relation to prior work: At this point it is possible to compare precisely the robust filtering problem discussed here with earlier formulations of robust filtering with a relative entropy tolerance considered in [1], [14], [15], and [18]. In [14], [15], Hansen and Sargent introduce the likelihood ratio function

$$M_t = \frac{\tilde{f}(X_{t+1}, Y_t)}{f(X_{t+1}, Y_t)} \quad (35)$$

between the distorted joint density of states $\{x_s, 0 \leq s \leq t+1\}$ and observations $\{y_s, 0 \leq s \leq t\}$ and their nominal joint density. Setting

$$M_{-1} = \frac{\tilde{f}_0(x_0)}{f_0(x_0)}$$

if $\mathcal{Z}_t$ denotes the σ -field generated by $\{z_s, 0 \leq s \leq t\}$ and x_0 , we have

$$E[M_t | \mathcal{Z}_{t-1}] = M_{t-1}$$

for $t \geq 0$, where $E[\cdot]$ denotes the expectation with respect to the nominal model distribution, so M_t is a martingale. Then the

relative entropy between the perturbed and nominal models over interval $0 \leq t \leq T$ can be expressed as

$$D(\tilde{f}, f) = E[M_T \ln M_T]. \quad (36)$$

Let

$$J = \sum_{t=0}^T \tilde{E} [\|x_t - \hat{x}_t\|^2] = \sum_{t=0}^T E [M_t \|x_t - \hat{x}_t\|^2] \quad (37)$$

denote the sum of the mean square estimation errors over interval $[0, T]$, where the estimate $\hat{x}_t = g_t(Y_t)$ depends causally on the past and current observations. If $g = (g_t, t \geq 0)$, the robust Kalman filtering problem of [14], [15] and the robust causal Wiener filtering problem of [1, Sec. 4] can be written as

$$\min_g \max_{\tilde{f}} J \quad (38)$$

where $\tilde{f}$ satisfies the constraint

$$D(\tilde{f}, f) = E[M_T \ln M_T] \leq c. \quad (39)$$

Since the causal Wiener filtering problem of [1, Sec. 4] is treated as a constrained static estimation problem, it does not require any additional structure. On the other hand for the robust Kalman filtering problem of [14], [15], the minimax problem (38) is formulated dynamically by introducing the martingale increments

$$m_t = \begin{cases} \frac{M_t}{M_{t-1}} & \text{if } M_{t-1} > 0 \\ 1 & \text{if } M_{t-1} = 0 \end{cases} \quad (40)$$

which can be expressed here as

$$m_t = \frac{\tilde{\phi}(z_t|x_t)}{\phi(z_t|x_t)}.$$

Then the chain rule (26) takes the form

$$E[M_T \ln M_T] = \sum_{t=0}^T E [M_{t-1} E[m_t \ln m_t | \mathcal{Z}_{t-1}]] + E[M_{-1} \ln M_{-1}] \quad (41)$$

and the dynamic minimax problem considered in [14] and [15] can be expressed as

$$\min_{g_t, t \geq 1} \max_{m_t, t \geq 1} J \quad (42)$$

where g_t and $m_t \geq 0$ are adapted, respectively, to $\mathcal{Y}_t$ (the sigma field spanned by Y_t) and $\mathcal{Z}_t$. Comparing (33) and (42), we see that (33) represents just a local or incremental version of minimax problem (42). In this respect it is worth pointing out that by integrating the incremental relative entropy constraint (28) with respect to the least favorable density $\tilde{f}_{t-1}(Y_{t-1})$ of Y_{t-1} and summing over all $0 \leq t \leq T$, we find that cumulatively, the incremental constraints (28) imply

$$D(\tilde{f}, f) \leq \sum_{t=0}^T c_t + D(\tilde{f}_0, f_0) \quad (43)$$

which has the form (39). In other words, the incremental constraint (28) just represents a way of dividing the relative entropy

tolerance budget c in separate portions c_t allocated to the distortion of the Markov model transition at each time step.

Formulation without commitment: The commitment assumption can be removed from the incremental formulation of robust filtering described above by adopting the conceptual framework of Hansen and Sargent in [14, Chap. 18] and [16], [24] for robust filtering without commitment. The key idea is, at time t , to apply distortions to both the transition dynamics $\phi_t(z_t|x_t)$ and the least-favorable *a priori* distribution $\tilde{f}_t(x_t|Y_{t-1})$. In [16] this is accomplished by introducing two distortion operators with constant risk-sensitivity parameters θ_1 and θ_2 , acting respectively on the transition dynamics and on the *a priori* density at time t based on the prior distortions and observations. Here, if $\tilde{f}_t(x_t|Y_{t-1})$ denotes a distorted version of $\tilde{f}_t(x_t|Y_{t-1})$, in addition to the constraint (28) for transition dynamics distortion, we could impose a constraint of the form

$$D(\tilde{f}_t, \tilde{f}_t) \leq d_t \quad (44)$$

on the allowed distortion of the least favorable conditional distortion for x_t based on the past observations Y_{t-1} . Then in the minimax problem (30) the maximization can be performed jointly over pairs $(\tilde{\phi}_t, \tilde{f}_t)$ satisfying constraints (28) and (44). Since the two constraints are convex, the structure of the resulting minimax problem is similar to (30). The main difficulty is algorithmic. The inner maximization introduces two Lagrange multipliers which need to be selected such that Karush–Kuhn–Tucker conditions are satisfied. The computation of the Lagrange multipliers seems rather difficult, in contrast to the case of a single Lagrange multiplier arising from the formulation of robust filtering with commitment. So we focus here on the case with commitment, leaving open the possibility that an implementable algorithm might be developed later for robust filtering without commitment under incremental constraints (28) and (44).

IV. ROBUST MINIMAX FILTER

The solution of the dynamic game (30) relies on extending Lemma 1 of [1] to the dynamic case. We start by observing that the objective function $J_t(\tilde{\phi}_t, g_t)$ specified by (31) is quadratic in g_t , and thus convex, and linear in $\tilde{\phi}_t$ and thus concave. The set $\mathcal{B}_t$ is convex and compact. Similarly, $\mathcal{G}_t$ is convex. It can also be made compact by requiring that the second moment of estimators $g_t \in \mathcal{G}_t$ should have a fixed but large upper bound. Then by Von Neumann's minimax theorem [25, p. 319], there exists a saddle point $(\tilde{\phi}_t^0, g_t^0)$ such that

$$J_t(\tilde{\phi}_t, g_t^0) \leq J_t(\tilde{\phi}_t^0, g_t^0) \leq J_t(\tilde{\phi}_t^0, g_t). \quad (45)$$

The real challenge is, however, not to establish the existence of a saddle point, but to characterize it completely. The second inequality in (45) implies that estimator g_t^0 is the conditional mean of x_{t+1} given Y_t based on the least-favorable density

$$\tilde{f}_{t+1}(x_{t+1}|Y_t) = \frac{\int \tilde{\phi}_t^0(z_t|x_t) \tilde{f}_t(x_t|Y_{t-1}) dx_t}{\int \int \tilde{\phi}_t^0(z_t|x_t) \tilde{f}_t(x_t|Y_{t-1}) dx_{t+1} dx_t} \quad (46)$$

obtained by marginalization and application of Bayes' rule to the least-favorable joint density $\tilde{\phi}_t^0(z_t|x_t) \tilde{f}_t(x_t|Y_{t-1})$ of

$(z_t, x_t) = (x_{t+1}, y_t, x_t)$ given Y_{t-1} . The robust estimator is then given by

$$\begin{aligned}\hat{x}_{t+1} &= g_t^0(y_t) = \tilde{E}[x_{t+1}|Y_t] \\ &= \int x_{t+1} \tilde{f}_{t+1}(x_{t+1}|Y_t) dx_{t+1}.\end{aligned}\quad (47)$$

Together, the conditional density evaluation (46) and expectation (47) implement the second inequality of saddle point identity (45). Let us turn now to the first inequality. For a fixed estimator g_t^0 , it requires finding the joint transition and observation density $\tilde{\phi}_t^0(z_t|x_t)$ maximizing $J_t(\tilde{\phi}_t, g_t^0)$ under the divergence constraint (28). The solution of this problem takes the following form.

Lemma 1: For a fixed estimator $g_t \in \mathcal{G}_t$, the function $\tilde{\phi}_t^0$ maximizing $J_t(\tilde{\phi}_t, g_t)$ under constraints (28) and (34) is given by

$$\tilde{\phi}_t^0(z_t|x_t) = \frac{1}{M_t(\lambda_t)} \exp\left(\frac{1}{2\lambda_t} \|x_{t+1} - g_t(y_t)\|^2\right) \phi_t(z_t|x_t). \quad (48)$$

In this expression, the normalizing constant $M_t(\lambda_t)$ is selected such that (34) holds. Furthermore, given a tolerance $c_t > 0$, there exists a unique Lagrange multiplier $\lambda_t > 0$ such that

$$D_t(\tilde{\phi}_t^0, \phi_t) = c_t. \quad (49)$$

Proof: For a given g_t , the function $J_t(\tilde{\phi}_t, g_t)$ is linear in $\tilde{\phi}_t$ and thus concave over the closed convex set $\mathcal{B}_t$, so it admits a unique maximum in $\mathcal{B}_t$. Because of the linearity of J_t with respect to $\tilde{\phi}_t$, this maximum is in fact located on the boundary of $\mathcal{B}_t$. To find the maximum, consider the Lagrangian

$$\begin{aligned}L_t(\tilde{\phi}_t, \lambda_t, \mu_t) &= J_t(\tilde{\phi}_t, g) + \lambda_t (c_t - D_t(\tilde{\phi}_t, \phi_t)) \\ &\quad + \mu_t (1 - I_t(\tilde{\phi}_t))\end{aligned}\quad (50)$$

where the Lagrange multipliers $\lambda_t \geq 0$ and μ_t are associated to inequality constraint (28) and equality constraint (34), respectively. We do not require explicitly that $\tilde{\phi}_t(z_t|x_t)$ should be non-negative, since the form (48) of the maximizing solution indicates that this constraint is satisfied automatically.

Then the Gateaux derivative [26, p. 17] of L_t with respect to $\tilde{\phi}_t$ in the direction of an arbitrary function u is given by

$$\begin{aligned}\nabla_{\tilde{\phi}_t, u} L_t(\tilde{\phi}_t, \lambda_t, \mu_t) &= \lim_{h \rightarrow 0} \frac{1}{h} [L_t(\tilde{\phi}_t + hu, \lambda_t, \mu_t) - L_t(\tilde{\phi}_t, \lambda_t, \mu_t)] \\ &= \iint \left[\frac{1}{2} \|x_{t+1} - g_t(y_t)\|^2 - (\lambda_t + \mu_t) - \lambda_t \ln \left(\frac{\tilde{\phi}_t}{\phi_t} \right) \right] \\ &\quad \times u(z_t, x_t) \tilde{f}_t(x_t|Y_{t-1}) dz_t dx_t.\end{aligned}\quad (51)$$

The Lagrangian is maximized by setting $\nabla_{\tilde{\phi}_t, u} L_t(\tilde{\phi}_t, \lambda_t, \mu_t) = 0$ for all functions u . Assuming $\lambda_t > 0$, this gives

$$\ln \left(\frac{\tilde{\phi}_t}{\phi_t} \right) = \frac{1}{2\lambda_t} \|x_{t+1} - g_t(y_t)\|^2 - \ln M_t \quad (52)$$

where

$$\ln M_t \triangleq 1 + \frac{\mu_t}{\lambda_t}.$$

Exponentiating (52) gives (48), where to ensure that normalization (34) holds, we must select

$$\begin{aligned}M_t(\lambda_t) &= \iint \exp\left(\frac{1}{2\lambda_t} \|x_{t+1} - g_t(y_t)\|^2\right) \phi_t(z_t|x_t) \\ &\quad \times \tilde{f}_t(x_t|Y_{t-1}) dz_t dx_t.\end{aligned}\quad (53)$$

At this point, all what is left is finding a Lagrange multiplier λ_t such that the solution $\tilde{\phi}_t^0$ given by (48) satisfies the KKT condition

$$\lambda_t (c_t - D_t(\tilde{\phi}_t^0, \phi_t)) = 0.$$

Since we already know that the maximizing $\tilde{\phi}_t^0$ is on the boundary of $\mathcal{B}_t$, the Lagrange multiplier $\lambda_t > 0$, so the KKT condition reduces to (49). By substituting (48) inside expression (28) for $D(\tilde{\phi}_t^0, \phi_t)$, we find

$$D(\tilde{\phi}_t^0, \phi_t) = \frac{1}{\lambda_t} J_t(\tilde{\phi}_t^0, g_t) - \ln(M_t(\lambda_t)). \quad (54)$$

Differentiating $\ln M_t(\lambda_t)$ gives

$$\frac{d}{d\lambda_t} \ln M_t(\lambda_t) = -\frac{1}{\lambda_t^2} J_t(\tilde{\phi}_t^0, g_t) \quad (55)$$

so that

$$\gamma(\lambda_t) \triangleq D(\tilde{\phi}_t^0, \phi_t) = -\lambda_t \frac{d}{d\lambda_t} \ln M_t(\lambda_t) - \ln(M_t(\lambda_t)). \quad (56)$$

The derivative of $\gamma(\lambda_t)$ is given by

$$\begin{aligned}\frac{d\gamma_t}{d\lambda_t} &= -\lambda_t \frac{d^2}{d\lambda_t^2} \ln M_t - 2 \frac{d}{d\lambda_t} \ln M_t \\ &= -\frac{1}{\lambda_t} \left[\frac{d}{d\lambda_t} \left(\lambda_t^2 \frac{d}{d\lambda_t} \ln M_t \right) \right] = \frac{1}{\lambda_t} \frac{d}{d\lambda_t} J_t(\tilde{\phi}_t^0, g_t) \\ &= -\frac{1}{4\lambda_t^3} \tilde{E} \left[\left(\|x_{t+1} - g_t(y_t)\|^2 \right. \right. \\ &\quad \left. \left. - \tilde{E} \left[\|x_{t+1} - g_t(y_t)\|^2 | Y_{t-1} \right] \right)^2 | Y_{t-1} \right] \\ &< 0\end{aligned}\quad (57)$$

so that $\gamma(\lambda_t)$ is a monotone decreasing function of λ_t . As $\lambda_t \rightarrow \infty$ we have obviously $\tilde{\phi}_t^0 \rightarrow \phi_t$, so that $\gamma(\infty) = 0$. Thus, provided c_t is located in the range of γ_t , which is the case if c_t is sufficiently small, there exists a unique λ_t such that $\gamma_t(\lambda_t) = c_t$. $\square$

Note that Lemma 1 makes no assumption about the form of the nominal transition density $\phi_t(z_t|x_t)$ and *a priori* density $\tilde{f}_t(x_t|Y_{t-1})$. Without additional assumptions, it is difficult to characterize precisely the range of function γ_t . When both of these densities are Gaussian, it will be shown below that the range of γ_t is $\mathbb{R}^+$, so that any positive divergence tolerance c_t can be achieved. However, in practice the tolerance c_t needs to be rather small in order to ensure that the robust estimator is not overly conservative. At this point is also worth observing that

Lemma 1 is just a variation of Theorem 2.1 in [27, p. 38] which sought to construct the minimum discrimination density (i.e., the density minimizing the divergence) with respect to a nominal density under various moment constraints. Here we seek to maximize the moment $\tilde{E}[\|x_{t+1} - g_t(y_t)\|^2 | Y_{t-1}]$ under a divergence constraint. From an optimization point of view, the two problems are obviously similar, and in fact the functional form (48) of the solution is the same for both problems.

Up to this point we have made no assumption on either the nominal transition and observations density $\phi_t(z_t|x_t)$ and estimator g_t , and in the characterization of the saddle point solution $(\tilde{\phi}_t^0, g_t^0)$ provided by identities (47) and (48), the robust estimator g_t^0 depends on least-favorable transition function $\tilde{\phi}_t^0$, and the least favorable transition density $\tilde{\phi}_t^0$ depends on robust estimator g_t^0 . This type of deadlock is typical of saddle point analyses, and to break it, we introduce now the assumption that $\phi_t(z_t|x_t)$ admits the Gaussian form (22) where as indicated earlier, the covariance matrix $K_{z_t|x_t}$ is positive definite, and we assume also that at time t the *a priori* conditional density

$$\tilde{f}_t(x_t|Y_{t-1}) \sim \mathcal{N}(\hat{x}_t, V_t). \quad (58)$$

Then, observe that the distortion term $\exp(\|x_{t+1} - g_t(y_t)\|^2/(2\lambda_t))$ appearing in expression (48) for the least-favorable transition function $\tilde{\phi}_t^0(z_t|x_t)$ depends only on z_t , but not x_t . Accordingly, if we introduce the marginal densities

$$\bar{f}_t(z_t|Y_{t-1}) = \int \phi_t(z_t|x_t) \tilde{f}_t(x_t|Y_{t-1}) dx_t \quad (59)$$

$$\tilde{f}_t^0(z_t|Y_{t-1}) = \int \tilde{\phi}_t^0(z_t|x_t) \tilde{f}_t(x_t|Y_{t-1}) dx_t \quad (60)$$

the density $\bar{f}_t(z_t|Y_{t-1})$ can be viewed as the pseudo-nominal density of $z_t = (x_{t+1}, y_t)$ conditioned on Y_{t-1} computed from the conditional least favorable density $\tilde{f}_t(x_t|Y_{t-1})$ and nominal transition density $\phi_t(z_t|x_t)$, and from (48) we obtain

$$\tilde{f}_t^0(z_t|Y_{t-1}) = \frac{1}{M(\lambda_t)} \exp\left(\frac{1}{2\lambda_t} \|x_{t+1} - g_t(y_t)\|^2\right) \bar{f}_t(z_t|Y_{t-1}). \quad (61)$$

Since densities $\phi_t(z_t|x_t)$ and $\tilde{f}_t(x_t|Y_{t-1})$ are both Gaussian, the integration (59) yields a Gaussian pseudo-nominal density

$$\bar{f}_t(z_t|Y_{t-1}) \sim \mathcal{N}\left(\begin{bmatrix} A_t \\ C_t \end{bmatrix} \hat{x}_t, K_{z_t}\right) \quad (62)$$

where the conditional covariance matrix K_{z_t} is given by

$$K_{z_t} = \begin{bmatrix} A_t \\ C_t \end{bmatrix} V_t \begin{bmatrix} A_t^T & C_t^T \end{bmatrix} + \begin{bmatrix} B_t \\ D_t \end{bmatrix} \begin{bmatrix} B_t^T & D_t^T \end{bmatrix}. \quad (63)$$

Integrating out x_t in (53) gives

$$M_t(\lambda_t) = \int \exp\left(\frac{1}{2\lambda_t} \|x_{t+1} - g_0(y_t)\|^2\right) \bar{f}_t(z_t|Y_{t-1}) dz_t$$

which ensures that $\tilde{f}_t^0(z_t|Y_{t-1})$ is a probability density. Furthermore, by direct substitution, we have

$$D(\tilde{f}_t^0, \bar{f}_t) = D(\tilde{\phi}_t^0, \phi_t) = c_t. \quad (64)$$

The least-favorable density $\tilde{f}_{t+1}(x_{t+1}|Y_t)$ specified by (46) can also be expressed in terms of $\tilde{f}_t^0(z_t|Y_{t-1})$ as

$$\tilde{f}_{t+1}(x_{t+1}|Y_t) = \frac{\tilde{f}_t^0(z_t|Y_{t-1})}{\int \tilde{f}_t^0(z_t|Y_{t-1}) dx_{t+1}}. \quad (65)$$

Equivalent Static Problem: Let

$$\bar{B}_t = \left\{ \tilde{f}_t : D(\tilde{f}_t, \bar{f}_t) \leq c_t \right\} \quad (66)$$

denote the ball of distorted densities $\tilde{f}_t(z_t)$ within a divergence tolerance c_t of pseudo-nominal density $\bar{f}_t(z_t|Y_{t-1})$. To this ball we can of course attach a static minimax estimation problem

$$\min_{g_t \in \mathcal{G}_t} \max_{\tilde{f}_t \in \bar{B}_t} J_t(\tilde{f}_t, g_t). \quad (67)$$

The solution of this problem satisfies the saddle point inequality

$$J(\tilde{f}_t, g_t^0) \leq J(\tilde{f}_t^0, g_t^0) \leq J_t(\tilde{f}_t^0, g_t^0). \quad (68)$$

At this point, observe that if $(\tilde{\phi}_t^0, g_t^0)$ solves the dynamic minimax game (30), and if $\tilde{f}_t^0$ is given by (61) with $g_t = g_t^0$, where λ_t is selected such that constraint (64) is satisfied, then $(\tilde{f}_t^0, g_t^0)$ is a saddle point of the static problem (67). In other words, the marginalization operation (60) has the effect of mapping the solution of dynamic game (30) into a solution of the static estimation problem (67). Note indeed that the solution of the maximization problem formed by the first inequality of (68) is given by (61) with $g_t = g_t^0$. Similarly, since g_t^0 is the mean of the conditional density $\tilde{f}_{t+1}(x_{t+1}|Y_t)$ specified by (65), it obeys the second inequality of (68).

Since the pseudo-nominal density $\bar{f}_t(z_t|Y_{t-1})$ specifying the center of ball $\bar{B}_t$ is Gaussian, Theorem 1 is applicable with $f \rightarrow \bar{f}_t$, $\tilde{f}_0 \rightarrow \tilde{f}_t^0$ and $g_0 \rightarrow g_t^0$. Hence the least-favorable density takes the form

$$\tilde{f}_t^0(z_t|Y_{t-1}) \sim \mathcal{N}\left(\begin{bmatrix} A_t \\ C_t \end{bmatrix} \hat{x}_t, \tilde{K}_{z_t}\right) \quad (69)$$

where the covariance matrix

$$\tilde{K}_{z_t} = \begin{bmatrix} \tilde{K}_{x_{t+1}} & K_{x_{t+1}y_t} \\ K_{y_t x_{t+1}} & K_{y_t} \end{bmatrix}$$

is obtained by perturbing only the (1,1) block

$$K_{x_{t+1}} = A_t V_t A_t^T + B_t B_t^T$$

of the covariance matrix K_{z_t} given by (63). The robust estimator takes the form

$$\hat{x}_{t+1} = g_t^0(y_t) = A_t \hat{x}_t + G_t(y_t - C_t \hat{x}_t) \quad (70)$$

with the matrix gain

$$G_t = K_{x_{t+1}y_t} K_{y_t}^{-1} \\ = (A_t V_t C_t^T + B_t D_t^T) (C_t V_t C_t^T + D_t D_t^T)^{-1}. \quad (71)$$

The least-favorable covariance matrix $\tilde{K}_{x_{t+1}}$ can be evaluated as follows. Let

$$P_{t+1} = K_{x_{t+1}} - K_{x_{t+1}y_t} K_{y_t}^{-1} K_{y_t x_{t+1}} \\ = (A_t - G_t C_t) V_t (A_t - G_t C_t)^T \\ + (B_t - G_t D_t) (B_t - G_t D_t)^T \quad (72)$$

and

$$V_{t+1} = \tilde{K}_{x_{t+1}} - K_{x_{t+1}y_t} K_{y_t}^{-1} K_{y_t x_{t+1}} \quad (73)$$

denote the nominal and least-favorable conditional covariance matrices of x_{t+1} given $\mathcal{Y}_t$. Then

$$V_{t+1}^{-1} = P_{t+1}^{-1} - \lambda_t^{-1} I_n \quad (74)$$

where the Lagrange multiplier $\lambda_t > r(P_{t+1})$ is selected such that

$$\gamma_t(\lambda_t) = \frac{1}{2} \left[\text{tr} \left((I_n - \lambda_t^{-1} P_{t+1})^{-1} - I_n \right) \right. \\ \left. + \ln \det (I_n - \lambda_t^{-1} P_{t+1}) \right] = c_t \quad (75)$$

where as indicated in (17), $\gamma_t(\lambda_t)$ is monotone decreasing over $(r(P_{t+1}), \infty)$ and has for range $\mathbb{R}^+$. Thus, for any divergence tolerance $c_t > 0$, there exists a matching Lagrange multiplier $\lambda_t > r(P_{t+1})$.

Summary: The least-favorable conditional distribution of x_{t+1} given Y_t is given by

$$\tilde{f}_{t+1}(x_{t+1}|Y_t) \sim \mathcal{N}(\hat{x}_{t+1}, V_{t+1}) \quad (76)$$

where the estimate $\hat{x}_{t+1}$ is obtained by propagating the filter (70), (71) and the conditional covariance matrix V_{t+1} is obtained from (72) and (74), with the Lagrange multiplier λ_t specified by (75). By writing $\theta_t = \lambda_t^{-1}$, we recognize immediately that the robust filter is a form of risk-sensitive filter of the type discussed in [7], [28, Chap. 10]. However, there is a new twist in the sense that, whereas standard risk-sensitive filtering uses a fixed risk sensitivity parameter θ , here θ is time-dependent. Specifically, in classical risk-sensitive filters, θ is an exponentiation parameter appearing in the exponential of quadratic cost to be minimized. Similarly, in earlier works [1], [14], [15], [17], [18] relating risk sensitive filtering with minimax filtering with a relative entropy constraint, a single global relative entropy constraint is imposed, resulting in a single Lagrange multiplier/risk sensitivity parameter. Here each component $\phi_t(z_t|x_t)$ of the model has an associated relative entropy constraint (27), where the tolerance c_t varies in inverse proportion with the modeler's confidence in the model component. In this respect, even if the

state-space model (18), (19) is time-invariant (A , B , C and D are constant) and the tolerance $c_t = c$ is constant, the risk sensitivity parameter λ_t^{-1} will be generally time-varying. On the other hand, if we insist on holding $\theta = \lambda^{-1}$ constant, it means that the tolerance $c_t = \gamma_t(\lambda)$ is time varying since the covariance matrix P_{t+1} given by (72) depends on time.

V. LEAST-FAVORABLE MODEL

In last section, we derived the robust filter, which is of course the most important component of the solution of the minimax filtering problem. For simulation and performance evaluation purposes, it is also useful to construct the least favorable model corresponding to the optimum filter. Before proceeding, note that if $e_t = x_t - \hat{x}_t$ denotes the state estimation error, by subtracting (70) from the state dynamics (18) and taking into account expression (19) for the observations, the estimation error dynamics are given by

$$e_{t+1} = (A_t - G_t C_t) e_t + (B_t - G_t D_t) v_t \quad (77)$$

where in the nominal model, the driving noise v_t is independent of error $e_t = x_t - \hat{x}_t$, since $\hat{x}_t$ depends exclusively on observations $\{y(s), 0 \leq s \leq t-1\}$.

To find the least-favorable model, we use the characterization (48) where $g_t = g_t^0$ is given by the robust filter (70). This gives

$$\tilde{\phi}_t^0(z_t|x_t) = \frac{1}{M_t(\lambda_t)} \exp \left(\frac{\|e_{t+1}\|^2}{2\lambda_t} \right) \phi_t(z_t|x_t). \quad (78)$$

At this point, recall that $\tilde{\phi}_t^0(z_t|x_t)$ is an unnormalized density. Specifically, integrating it over z_t does not yield one, but as we shall see below, a positive function of $e_t = x_t - \hat{x}_t$. This feature indicates that the maximizing player has the opportunity to change retroactively the least-favorable density of x_t (and therefore of earlier states) by selecting the model component $\tilde{\phi}_t^0(z_t|x_t)$. Properly accounting for this retroactive change forms an important aspect of the derivation of the least-favorable model. Instead of attempting to characterize directly the least-favorable density of z_t given x_t , it is easier to find the least-favorable density of the driving noise

$$v_t = \Gamma_t^{-1} \left(z_t - \begin{bmatrix} A_t \\ C_t \end{bmatrix} x_t \right). \quad (79)$$

Given x_t , the transformation (79) establishes a one-to-one correspondence between z_t and v_t , so there is no loss of information in characterizing the least-favorable model in terms of v_t . Let $\psi_t(v_t)$ and $\tilde{\psi}_t(v_t|e_t)$ denote, respectively, the nominal and least-favorable densities of noise v_t , where as will be shown below, $\tilde{\psi}_t(v_t|e_t)$ actually depends on e_t . The nominal distribution is given by

$$\psi_t(v_t) = \frac{1}{(2\pi)^{n+p}} \exp \left(-\|v_t\|^2/2 \right). \quad (80)$$

Assume that we seek to construct the least-favorable model of z_t over a fixed interval $0 \leq t \leq T$, and that the least-favorable

noise distribution $\tilde{\psi}_s(v_s|e_s)$ has been identified for $t+1 \leq s \leq T$. Accordingly, we have

$$\prod_{s=t+1}^T \exp\left(\frac{\|e_{s+1}\|^2}{2\lambda_s}\right) \psi_s(v_s) \sim \exp\left(\frac{\|e_{t+1}\|_{\Omega_{t+1}^{-1}}^2}{2}\right) \prod_{s=t+1}^T \tilde{\psi}_s(v_s|e_s) \quad (81)$$

where $\sim$ indicates equality up to a multiplicative constant. The term $\exp(\|e_{t+1}\|_{\Omega_{t+1}^{-1}}^2/2)$ appearing in the above expression accounts for the cumulative effect of retroactive probability density changes performed by the maximizing player. Here Ω_t denotes a positive definite matrix of dimension n which is evaluated recursively. Then the least-favorable model $\tilde{\psi}_t(v_t|e_t)$ is obtained by backward induction. Decrementing the index t by 1 in (81) gives the identity

$$\exp\left(\frac{\left(\|e_{t+1}\|_{\Omega_{t+1}^{-1}}^2 + \lambda_t^{-1}\|e_{t+1}\|^2\right)}{2}\right) \psi_t(v_t) \sim \tilde{\psi}_t(v_t|e_t) \exp\left(\frac{\|e_t\|_{\Omega_t^{-1}}^2}{2}\right). \quad (82)$$

Let

$$W_{t+1} \triangleq (\Omega_{t+1}^{-1} + \lambda_t^{-1}I_n)^{-1}. \quad (83)$$

Then by substituting the error dynamics (77), the left-hand side of identity (82) becomes

$$\exp\left(\frac{\left(\|(A_t - G_t C_t)e_t + (B_t - G_t D_t)v_t\|_{W_{t+1}^{-1}}^2 - \|v_t\|^2\right)}{2}\right) \quad (84)$$

and the right-hand side of (82) is obtained by decomposing the quadratic exponent of (84) as a sum of squares in v_t and e_t . By doing so, we find that the least-favorable noise density is given by

$$\tilde{\psi}_t(v_t|e_t) \sim \mathcal{N}(H_t e_t, \tilde{K}_{v_t}) \quad (85)$$

where

$$\tilde{K}_{v_t} = (I_{n+p} - (B_t - G_t D_t)^T W_{t+1}^{-1} (B_t - G_t D_t))^{-1} \quad (86)$$

and

$$H_t = \tilde{K}_{v_t} (B_t - G_t D_t)^T W_{t+1}^{-1} (A_t - G_t C_t). \quad (87)$$

Thus, the least-favorable density of the noise v_t involves a perturbation of both the mean and the variance of the nominal noise distribution. The mean perturbation is proportional to the filtering error e_t , which creates a coupling between the robust filter and the least favorable model specified by dynamics and observations (18), (19) and least-favorable noise statistics (85)–(87).

Finally, by matching quadratic components in e_t on both sides of (82), we find

$$\Omega_t^{-1} = (A_t - G_t C_t)^T W_{t+1}^{-1} (A_t - G_t C_t) + H_t^T \tilde{K}_{v_t}^{-1} H_t \quad (88)$$

where $\tilde{K}_{v_t}$ and H_t are given by (86) and (87). By using the matrix inversion lemma [29, p. 48]

$$(\alpha + \beta\gamma\delta)^{-1} = \alpha^{-1} - \alpha^{-1}\beta(\gamma^{-1} + \delta\alpha^{-1}\beta)^{-1}\delta\alpha^{-1}$$

with $\alpha = W_{t+1}$, $\beta = (B_t - G_t D_t)$, $\gamma = -I_{n+p}$ and $\delta = (B_t - G_t D_t)^T$ on the right-hand side of (88), we obtain

$$\Omega_t^{-1} = (A_t - G_t C_t)^T [W_{t+1} - (B_t - G_t D_t)(B_t - G_t D_t)^T]^{-1} \times (A_t - G_t C_t). \quad (89)$$

The recursion (89), together with (83) specifies a backward recursion which is used to account for retroactive changes of previous least-favorable model densities performed by the maximizing player. The backwards recursion is initialized with $\Omega_{T+1}^{-1} = 0$, or equivalently,

$$W_{T+1} = \lambda_T I_{n+p}. \quad (90)$$

The least-favorable noise model (85)–(87) can be viewed as a time-varying version of the least-favorable model derived asymptotically for the case of a constant model by Hansen and Sargent in [14, Sec. 17.7]. Specifically, the dynamics of the least-favorable model described in [14] are expressed in terms of the solution of a deterministic infinite-horizon linear-quadratic regulator problem. The counterpart of this regulator is formed here by backward recursion (89), (83).

The model (85)–(87) indicates that the driving noise v_t admits the representation

$$v_t = H_t e_t + L_t \epsilon_t \quad (91)$$

where L_t is an arbitrary matrix square root of $\tilde{K}_{v_t}$, i.e.,

$$L_t L_t^T = [I_{n+p} - (B_t - G_t D_t)^T W_{t+1}^{-1} (B_t - G_t D_t)]^{-1} \quad (92)$$

and ϵ_t is a zero-mean WGN of variance I_{n+p} . Accordingly, as was previously observed in [14], if

$$\xi_t \triangleq \begin{bmatrix} x_t \\ e_t \end{bmatrix} \quad (93)$$

the least-favorable model admits a state-space representation

$$\begin{aligned} \xi_{t+1} &= \tilde{A}_t \xi_t + \tilde{B}_t \epsilon_t \\ y_t &= \tilde{C}_t \xi_t + \tilde{D}_t \epsilon_t \end{aligned} \quad (94)$$

with twice the dimension of the nominal state-space model, where

$$\begin{aligned} \tilde{A}_t &= \begin{bmatrix} A_t & B_t H_t \\ 0 & A_t - G_t C_t + (B_t - G_t D_t) H_t \end{bmatrix} \\ \tilde{B}_t &= \begin{bmatrix} B_t \\ (B_t - G_t D_t) \end{bmatrix} \\ \tilde{C}_t &= [C_t \quad D_t H_t], \quad \tilde{D}_t = D_t L_t. \end{aligned} \quad (95)$$

Note that the model (94), (95) is constructed by performing first a forward sweep of the risk-sensitive filter (71), (72) over interval $[0, T]$ to generate the gains G_t , followed by a backward sweep used to evaluate the matrix sequence W_t . Thus, the least-favorable model is constructed in a *nonsequential* manner, since increasing the simulation interval beyond $[0, T]$ requires performing a new backward sweep of recursion (89), (83).

The model (94) can be used to assess the performance of any estimation filter designed under the assumption that the nominal model (18), (19) is valid. Let G'_t be an arbitrary time-dependent gain sequence, and let $\hat{x}'_t$ be the state estimate generated by the recursion

$$\hat{x}'_{t+1} = A_t \hat{x}'_t + G'_t (y_t - C_t \hat{x}'_t). \quad (96)$$

Let $e'_t = x_t - \hat{x}'_t$ denote the corresponding filtering error. When the actual data is generated by the least-favorable model (94), (95), by subtracting recursion (96) from the first component of the state dynamics (94), we obtain

$$\begin{bmatrix} e'_{t+1} \\ e_{t+1} \end{bmatrix} = \left(\tilde{A}_t - \begin{bmatrix} G'_t \\ 0 \end{bmatrix} \tilde{C}_t \right) \begin{bmatrix} e'_t \\ e_t \end{bmatrix} + \left(\tilde{B}_t - \begin{bmatrix} G'_t \\ 0 \end{bmatrix} \tilde{D}_t \right) \epsilon_t. \quad (97)$$

The recursion (97) can be used to evaluate the performance of filter (96) when the data is generated by the least-favorable model (94), (95). Specifically, consider the covariance matrix

$$\Pi_t = \tilde{E} \begin{bmatrix} e'_t \\ e_t \end{bmatrix} \begin{bmatrix} (e'_t)^T & e_t^T \end{bmatrix}.$$

By using the dynamics (97) derived under the assumption that the data is generated by the least-favorable model, we obtain the Lyapunov equation

$$\begin{aligned} \Pi_{t+1} = & \left(\tilde{A}_t - \begin{bmatrix} G'_t \\ 0 \end{bmatrix} \tilde{C}_t \right) \Pi_t \left(\tilde{A}_t - \begin{bmatrix} G'_t \\ 0 \end{bmatrix} \tilde{C}_t \right)^T \\ & + \left(\tilde{B}_t - \begin{bmatrix} G'_t \\ 0 \end{bmatrix} \tilde{D}_t \right) \left(\tilde{B}_t - \begin{bmatrix} G'_t \\ 0 \end{bmatrix} \tilde{D}_t \right)^T \end{aligned} \quad (98)$$

which can be used to evaluate the performance of filter (97) when it is applied to the least-favorable model. For the special case where G'_t is the Kalman gain sequence, this yields the performance of the standard Kalman filter.

VI. SIMULATIONS

To illustrate the behavior of the robust filtering algorithm specified by (70)–(75), we consider a constant state-space model employed earlier in [11], [12]

$$\begin{aligned} A &= \begin{bmatrix} 0.9802 & 0.0196 \\ 0 & 0.9802 \end{bmatrix} \\ BB^T = Q &= \begin{bmatrix} 1.9608 & 0.0195 \\ 0.0195 & 1.9605 \end{bmatrix} \\ C &= [1 \quad -1], \quad DD^T = 1. \end{aligned}$$

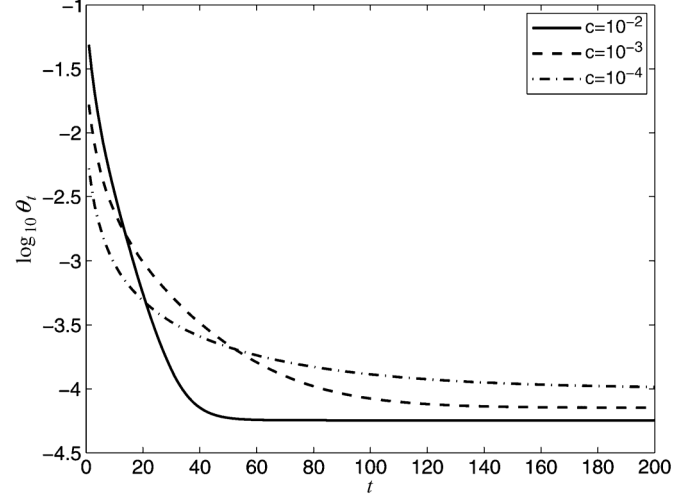


Fig. 1. Plot of time varying parameter $\theta_t = \lambda_t^{-1}$ (logarithmic scale) for $c = 10^{-2}$, 10^{-3} and 10^{-4} .

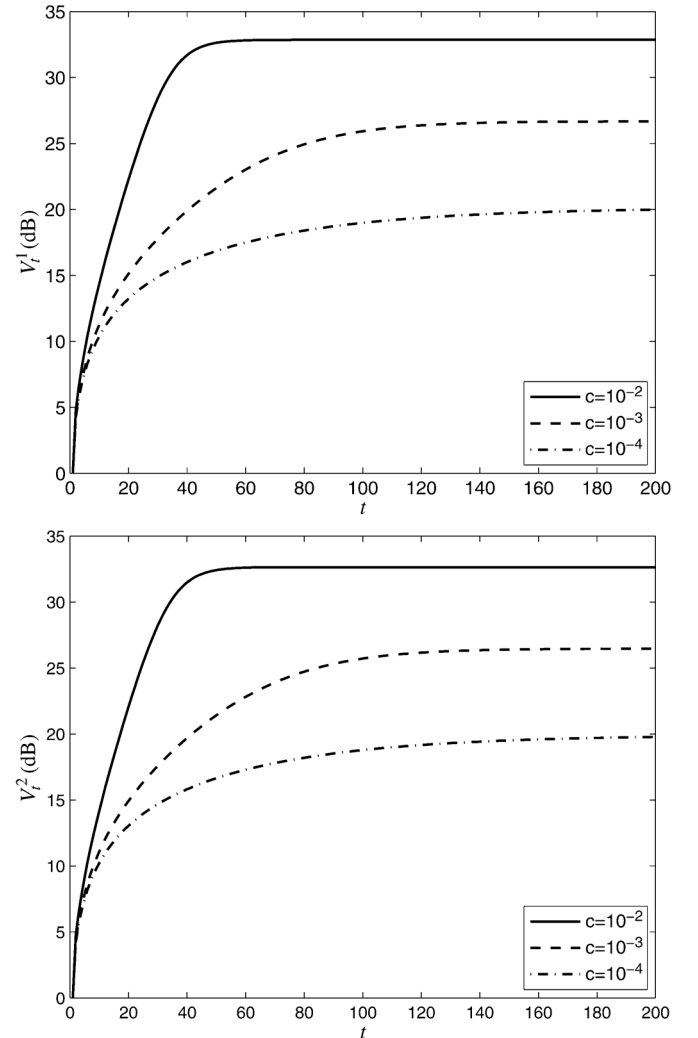


Fig. 2. Error variance of x_{1t} and x_{2t} (dB scale) for $c = 10^{-2}$, 10^{-3} , and 10^{-4} .

The nominal process noise Bv_t and measurement noise Dv_t are assumed to be uncorrelated, so that $BD^T = 0$, and the initial

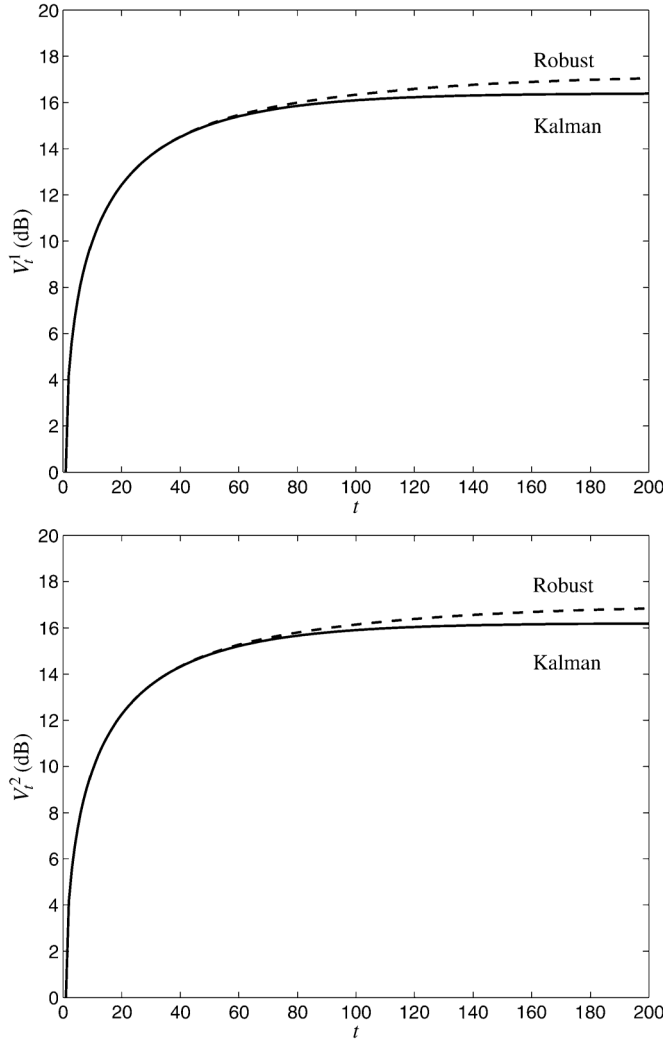


Fig. 3. Error variance of x_{1t} and x_{2t} (dB scale) when the risk-sensitive filter with $c = 10^{-4}$ and the Kalman filter are applied to the nominal model.

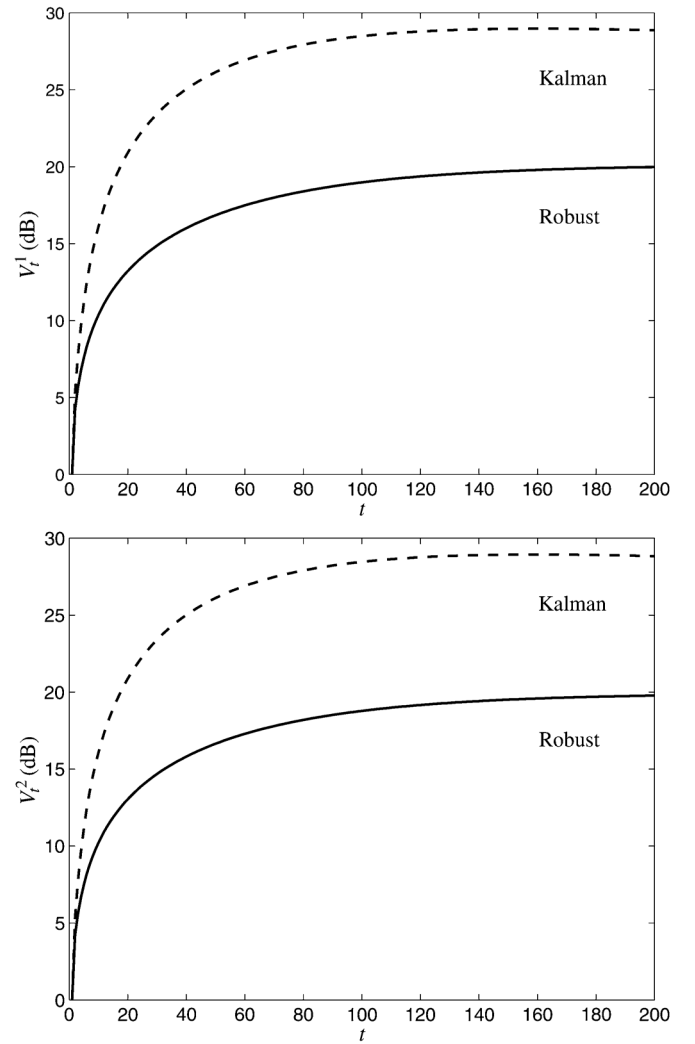


Fig. 4. Error variance of x_{1t} and x_{2t} (dB scale) when the risk-sensitive filter with $c = 10^{-4}$ and the Kalman filter are applied to the least-favorable model.

value of the least-favorable error covariance matrix is selected as

$$V_0 = I_2.$$

We apply the robust filtering algorithm over an interval of length $T = 200$ for progressively tighter values 10^{-2} , 10^{-3} and 10^{-4} of the relative entropy tolerance c . The corresponding time-varying risk-sensitivity parameters $\theta_t = \lambda_t^{-1}$ obtained from (75) are plotted in Fig. 1. The least-favorable variances (the (1,1) and (2,2) entries of V_t) of the two states are plotted as functions of time in Fig. 2. As can be seen from the plots, although the relative entropy tolerance bounds that we consider are small, increasing the tolerance c by a factor 10 leads to an increase of about 7 dB in the state error variances.

Next, for a tolerance $c = 10^{-4}$, we compare the performance of the risk-sensitive and Kalman filters for the nominal model, and for the least-favorable model constructed as indicated in Section V. The variances of the two-states for the nominal model are shown in Fig. 3. Clearly, the loss of performance

of the risk-sensitive filter compared to the Kalman filter is less than 1 dB.

On the other hand, as indicated in Fig. 4, when the risk-sensitive and Kalman filters are applied to the least-favorable model, the Kalman filter performance is about 8 dB worse than the robust filter. Note that to allow the backward recursion (83), (89) to reach steady state, the backward model is computed for a larger interval, and only the first 200 samples of the simulation interval are retained, since later samples are affected by transients of the least-favorable model.

An interesting aspect of the dynamics of the Riccati equation with a time-varying risk-sensitivity parameter $\theta_t = \lambda_t^{-1}$ is that the steady-state value for V_t does not depend exclusively on the final value of θ_t . To illustrate this observation, Fig. 5 compares the values of the diagonal elements of V_t for the incremental case examined here and for a standard risk-sensitive filter whose constant risk-sensitivity parameter $\theta = 10^{-4}$ matches the final value of θ_t in the incremental case. The limits for V_t are different, suggesting that the dynamics of the risk-sensitive Riccati equation with a time-varying risk-sensitivity parameter deserve further investigation.

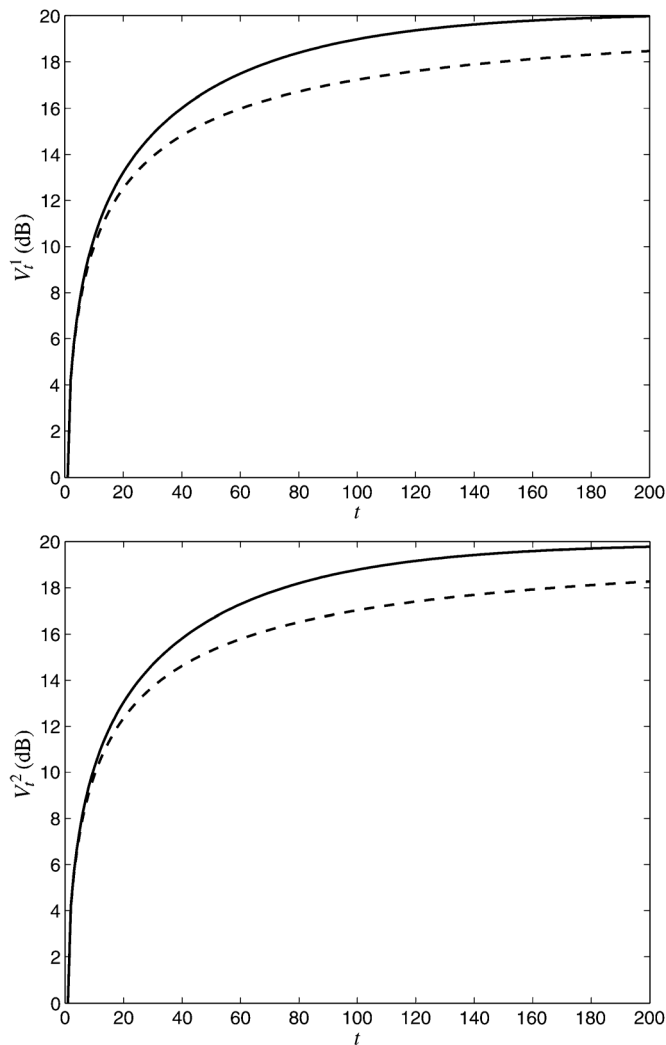


Fig. 5. Values of V_t^1 and V_t^2 for an incremental risk-sensitive (RS) filter with $c = 10^{-4}$ (solid line) and a conventional RS filter with $\theta = 10^{-4}$ (dashed line).

VII. CONCLUSION

In this paper, we have considered a robust state-space filtering problem with an incremental relative entropy constraint. The problem was formulated as a dynamic minimax game, and by extending results presented in [1], it was shown that the minimax filter is a risk-sensitive filter with a time varying risk-sensitive parameter. The associated least-favorable model was constructed by performing a backward recursion which keeps track of retroactive probability changes made by the maximizing player. The results obtained are similar in nature to those derived by Hansen and Sargent [14], [15] for the minimax problem (42) when a single relative entropy constraint is applied to the overall state-space model and the maximizing agent is required to operate under commitment.

A number of issues remain unresolved. For the case of a constant state-space model, it would be of interest to establish the convergence under appropriate conditions of the robust filtering recursions and of the backwards least-favorable model recursions. One also has to wonder if the results derived here for Gauss–Markov models could be extended to classes of systems, such as partially observed Markov chains, for which robust filtering with an overall relative entropy constraint was considered previously in [30].

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Bernard C. Levy (S'74–M'78–SM'90–F'94) received the diploma of Ingénieur Civil des Mines from the Ecole Nationale Supérieure des Mines, Paris, France, in 1974, and the Ph.D. degree in electrical engineering from Stanford University, Stanford, CA, in 1979.

From July 1979 to June 1987, he was Assistant and then Associate Professor in the Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology, Cambridge. Since July 1987, he has been with the University of California (UC), Davis, where he is Professor of Electrical Engineering and a member of the Graduate Group in Applied Mathematics. He served as Chair of the Department of Electrical and Computer Engineering at UC Davis from 1996 to 2000. He was a Visiting Scientist at the Institut de Recherche en Informatique et Systèmes Aléatoires (IRISA), Rennes, France, from January to July 1993, and at the Institut National de Recherche en Informatique et Automatique (INRIA), Rocquencourt, France, from September to December 2001. He is the author of the book *Principles of Signal Detection and Parameter Estimation* (Springer, 2008). His research interests include statistical signal processing, estimation, detection, and multidimensional signal processing.

Dr. Levy served as Associate Editor of the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I, of the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II, and of the *EURASIP Journal on Advances in Signal Processing*. He is currently an Associate Editor of *Signal Processing*. He is a member of SIAM.



Ramine Nikoukha received the Ph.D. degree from the Department of Electrical Engineering and Computer Science, Massachusetts Institute of Technology, Cambridge, in 1988.

He is currently Software Development Director at Altair Development France, on leave from his position as Directeur de Recherche at Inria Paris-Rocquencourt center. His current professional interests are in the areas of modeling and simulation of hybrid dynamical systems. He has also worked in the fields of systems, control and signal processing (in particular failure detection). He has published over 100 journal and conference papers, and coauthored four books. As a member of the original team that developed the scientific software package Scilab, he has contributed to a number of its toolboxes. He is in particular the creator of the modeling and simulation environment Scicos.